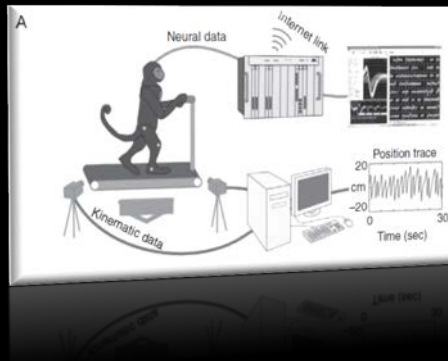
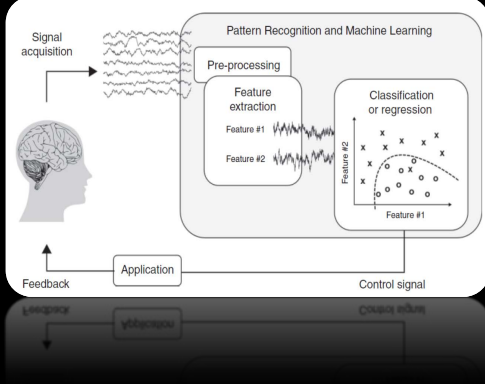


NATURAL COMPUTATION --- BRAIN COMPUTER INTERFACING

Minghui Shi

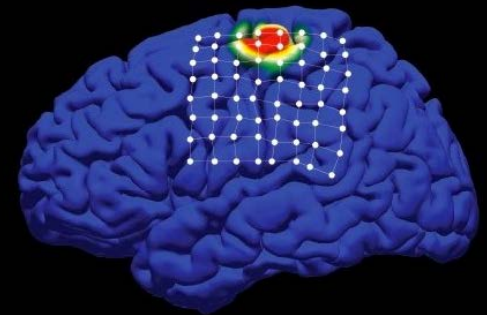
smh@xmu.edu.cn

15880281262

Xiamen University, China

Brain-Computer Interfacing

An Introduction



Rajesh P. N. Rao

CAMBRIDGE

上课时间、地点

□ 时间：（共32学时）

□ 1-16周 5-6节 （本学期共18周）

□ 地点：

□ 海韵教学楼 304

本讲目的

- 目的：
 - ▣ 整体上认识课程

本讲内容

- 课件下载、课程历史沿革、课程简介、培养目标
- **What is Natural Computation?**
- What's the relationship between **Natural Computation** and *Brain Computer Interface* ?
- **Characteristic of this course**
- **What is Brain Computer Interface?**
- **Textbook & Reference Book**
- **Content & Topics**
- **Basic Requirement**
- **Group Task**
- **Score Assessment**

课件下载

□ 课件下载: <http://course.xmu.edu.cn/>

课程历史沿革

□ 课程历史沿革

▣ 15162学期 全英文 本科生

▣ 16172学期 全英文 本科生

▣ 17182学期 全英文 本科生

▣ 19202学期 双语 本科生

▣ 18191学期 全英文 研究生（主要面向国际生）

▣ 19201学期 全英文 研究生（主要面向国际生）

课程简介

- 《自然计算》属于高年级本科生学科类方向性课程。
- 采用**双语教学**。（英文教材和英文课件为主，中文教材为参考）
- 自然计算涉及的议题广泛。课程以脑机接口这一近年兴起的自然计算技术为主题，系统地介绍脑与外界直接沟通交流的原理和方法。
- 具体地说，即介绍如何通过提取与分析脑信号，使生物（人或动物）不用通过五官与四肢，而凭借脑直接与外界交换信息的原理和方法，并讨论其具体的应用及相关的社会与伦理议题，旨在使学生对脑机接口这一新型自然计算技术有较为全面的了解。

培养目标

- 一方面，培养或提高学生研读英文文献的能力。
- 另一方面，较为全面地介绍脑机接口这一新兴的自然计算技术，基本内容包括：
 - ▣ 神经科学背景理论
 - ▣ 脑信号的提取与分析方法
 - ▣ 脑机接口技术的原理与应用
 - ▣ 相关的社会伦理议题

Congratulations!

- From this course, you are expected to not only **improve your English**, but also learn an amazing modern technology ---- **Brain Computer Interface**, one kind of advanced **Natural Computation** technologies, including basic principles of **Neuroscience, Signal Processing, Machine Learning**, etc. and various novel **BCI Applications**.

What is Natural Computation?

- A: From Wikipedia
(https://en.wikipedia.org/wiki/Natural_computing) :
- Natural computing, also called natural computation, is a terminology introduced to encompass three classes of methods:
 - ▣ 1) those that take inspiration from nature for the development of novel problem-solving techniques;
 - ▣ 2) those that are based on the use of computers to synthesize natural phenomena;
 - ▣ 3) those that employ natural materials (e.g., molecules) to compute.
- The main fields of research that compose these three branches are *artificial neural networks, evolutionary algorithms, swarm intelligence, artificial immune systems, fractal geometry, artificial life, DNA computing, and quantum computing*, among others.

What's the relationship between **Natural Computation** and ***Brain Computer Interface*** ?

- Now, we know what **Natural Computation** is.
- ***Brain Computer Interface*** is the topic of our course.
- You may wonder what's the relationship between **Natural Computation** and ***Brain Computer Interface***.
 - ◆ Although ***Brain Computer Interface*** is not listed in the web above, it is a kind of **Natural Computation** technology.
 - ◆ ***Brain Computer Interface*** is one of the important branches of **Natural Computation**.
 - ◆ Furthermore, ***Brain Computer Interface*** is one of **Natural Computation** technologies with rapid development in recent years.

Characteristic of this course

- This course has the following characteristics:
- **English**
 - ▣ Using English throughout the course.
- **Advanced technique**
 - ▣ Studying on various aspects of Brain-Computer Interface.
- **Developing capability of research**
 - ▣ Discussing and learning many references.
- **Interesting**
 - ▣ Many references, videos and applications will be discussed.

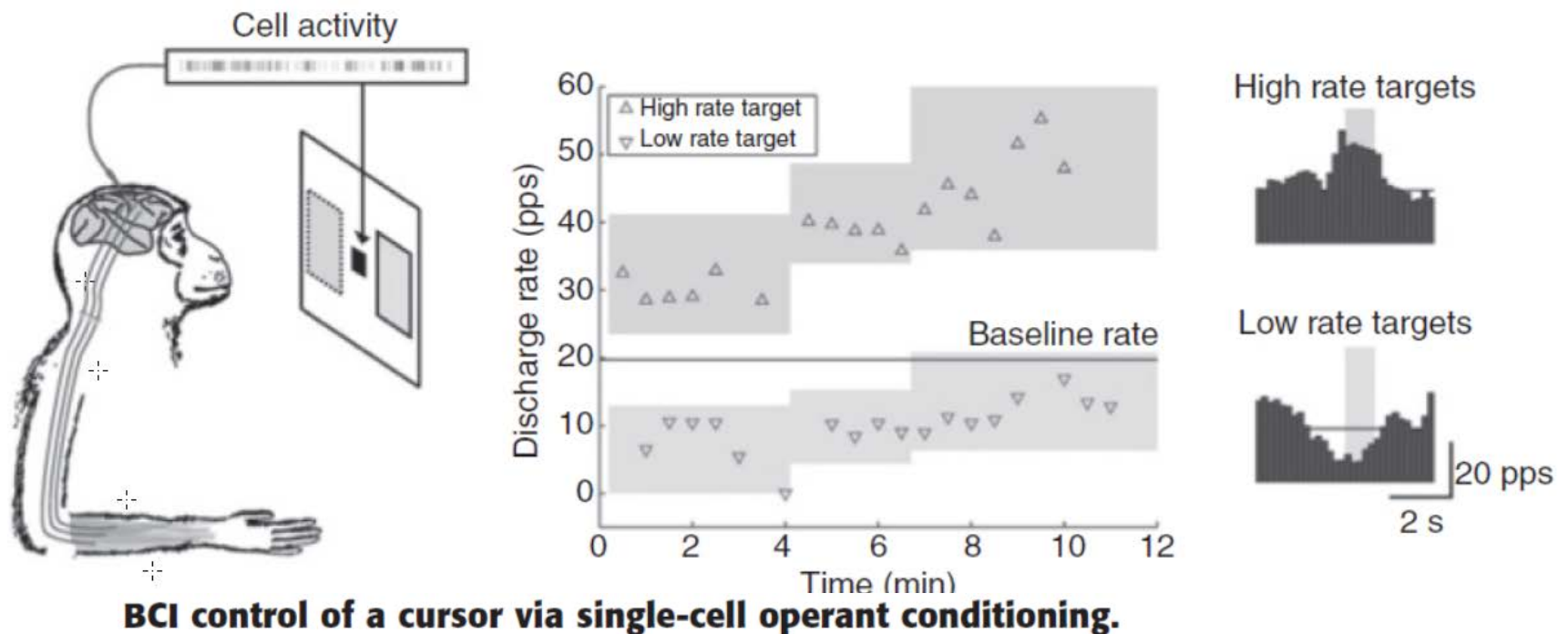
What is Brain Computer Interface?

- At present, it may be hard for you to totally understand ***Brain Computer Interface (BCI)***.
- To allow you have an impression on it, let us watch some videos and pictures related to BCI and our course.

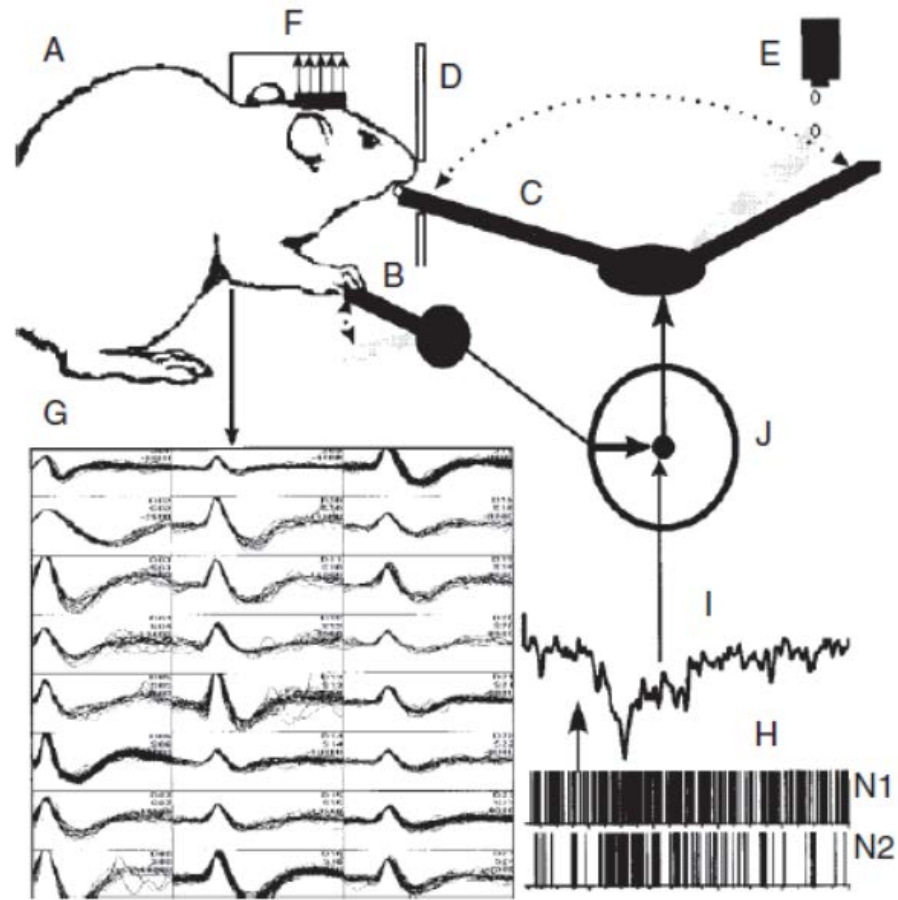
BCI Videos

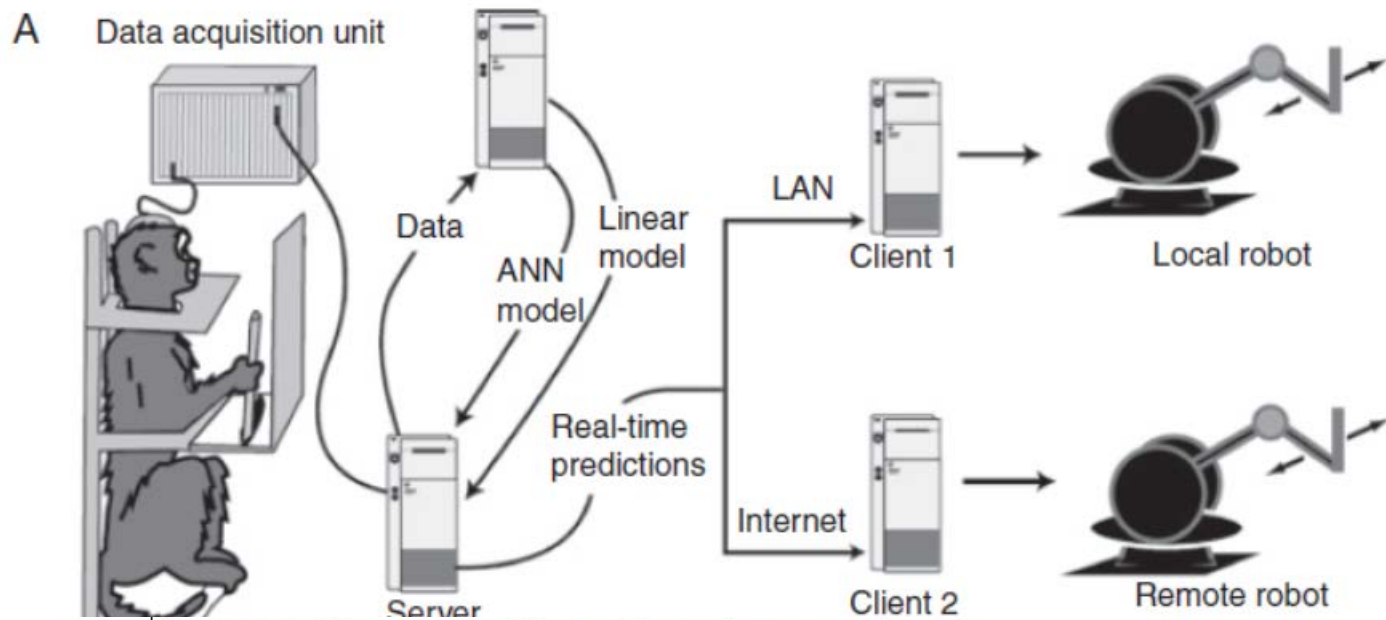
- http://open.163.com/movie/2009/11/V/A/M6I12OHDE_M6I134KVA.html
- http://www.soku.com/search_video/q_brain%20computer%20interface?f=1&kb=0402000000000000__brain%20computer%20interface&_rp=1451568674644XzYhjl&_rp=1451568674644XzYhjl
- <http://www.youku.com/> **search: brain computer interface**
- http://v.youku.com/v_show/id_XMjMwMTQwNjQ0.html?from=s1.8-1-1.2
- http://v.youku.com/v_show/id_XMTM4NzE2MjAwMA==.html?from=s1.8-1-1.2
- http://v.youku.com/v_show/id_XODA0NTAyMjA4.html?from=s1.8-1-1.2
- http://v.youku.com/v_show/id_XMTM4NzE5Njg0MA==.html?from=s1.8-1-1.2

BCI Pictures From Textbook

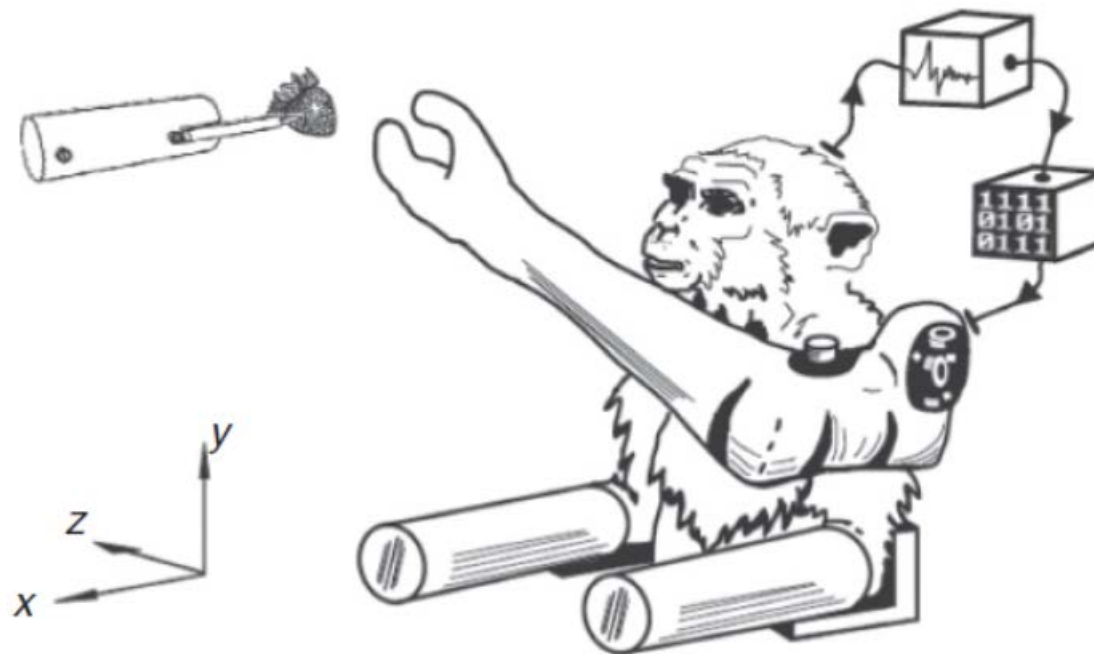


Invasive BCI in rats.

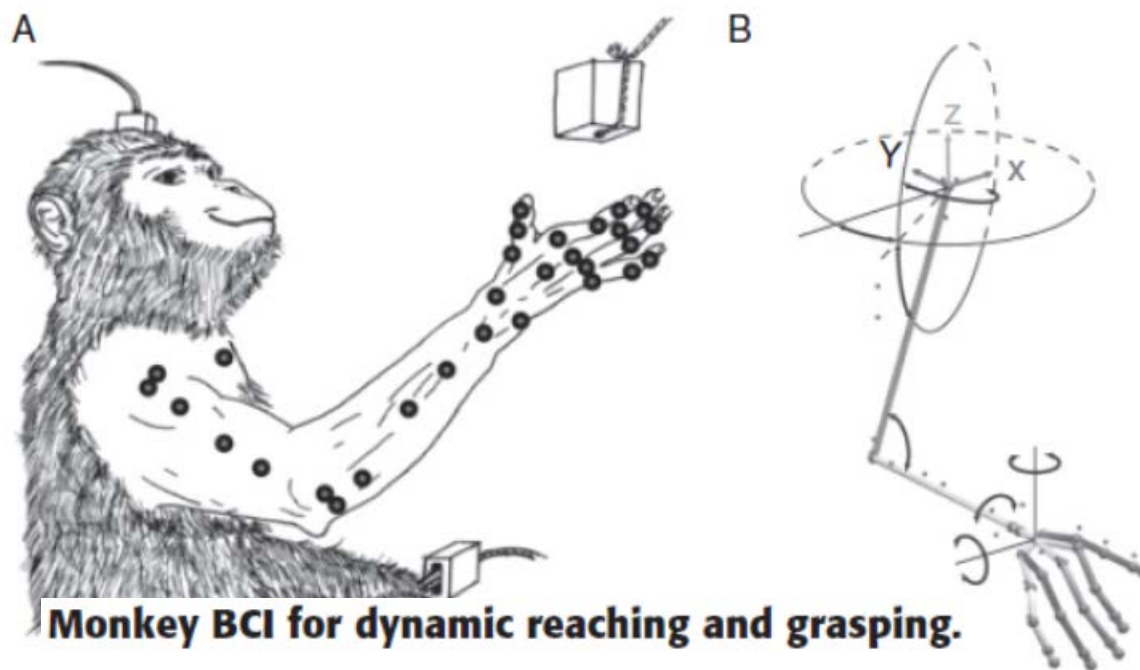




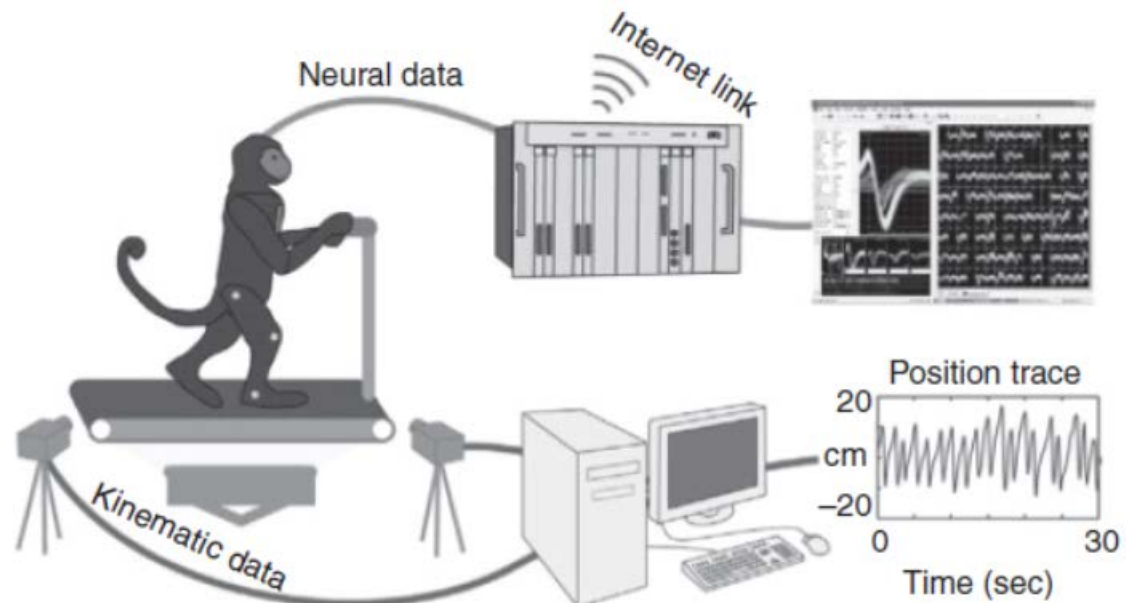
Monkey BCI for a one-dimensional control task.



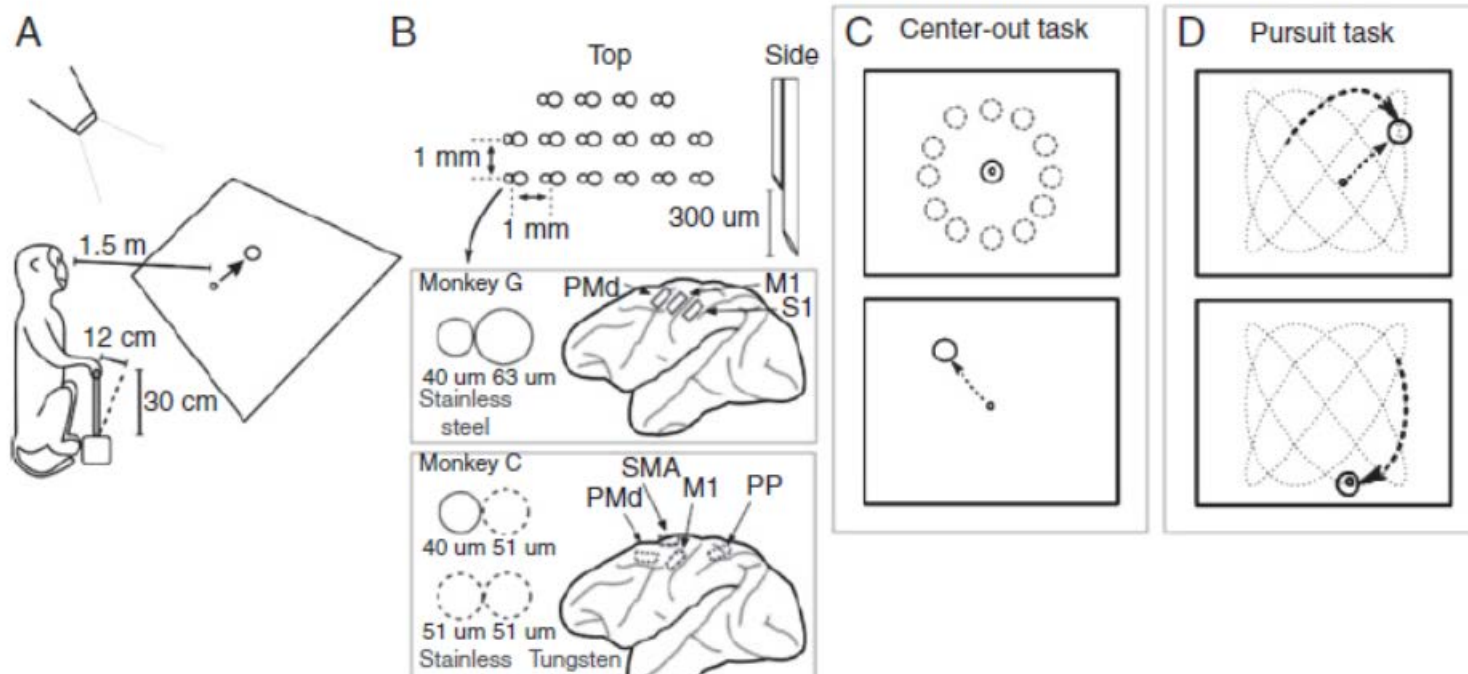
BCI control of a prosthetic arm and gripper for self-feeding.



A

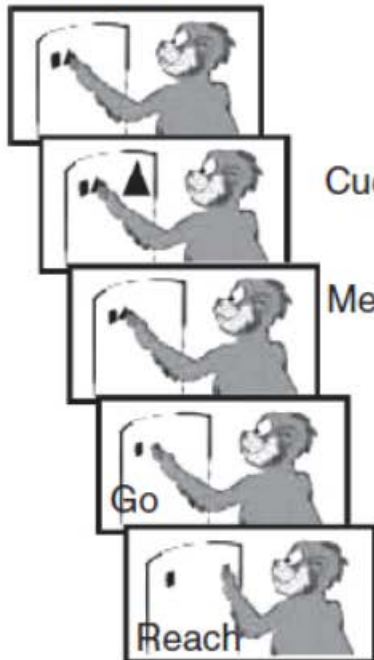


Predicting lower-limb kinematics using neural activity.

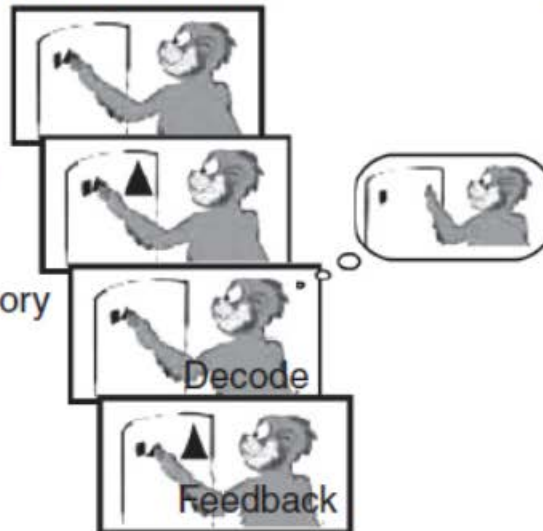


Experimental setup for demonstrating BCI control of a cursor.

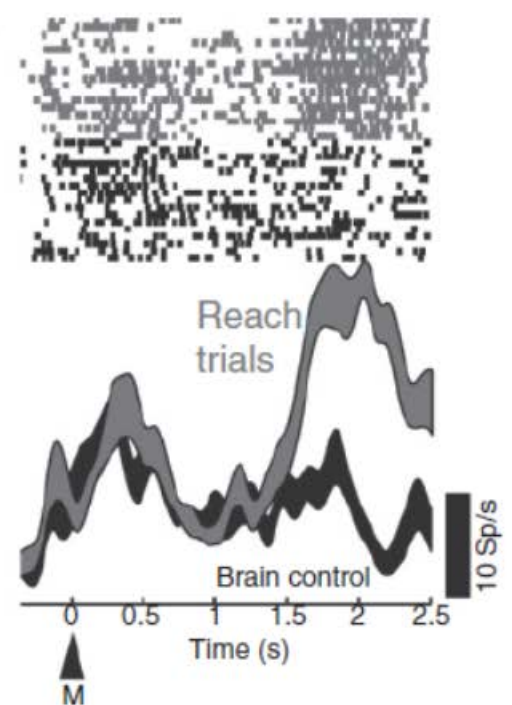
A Reach trial



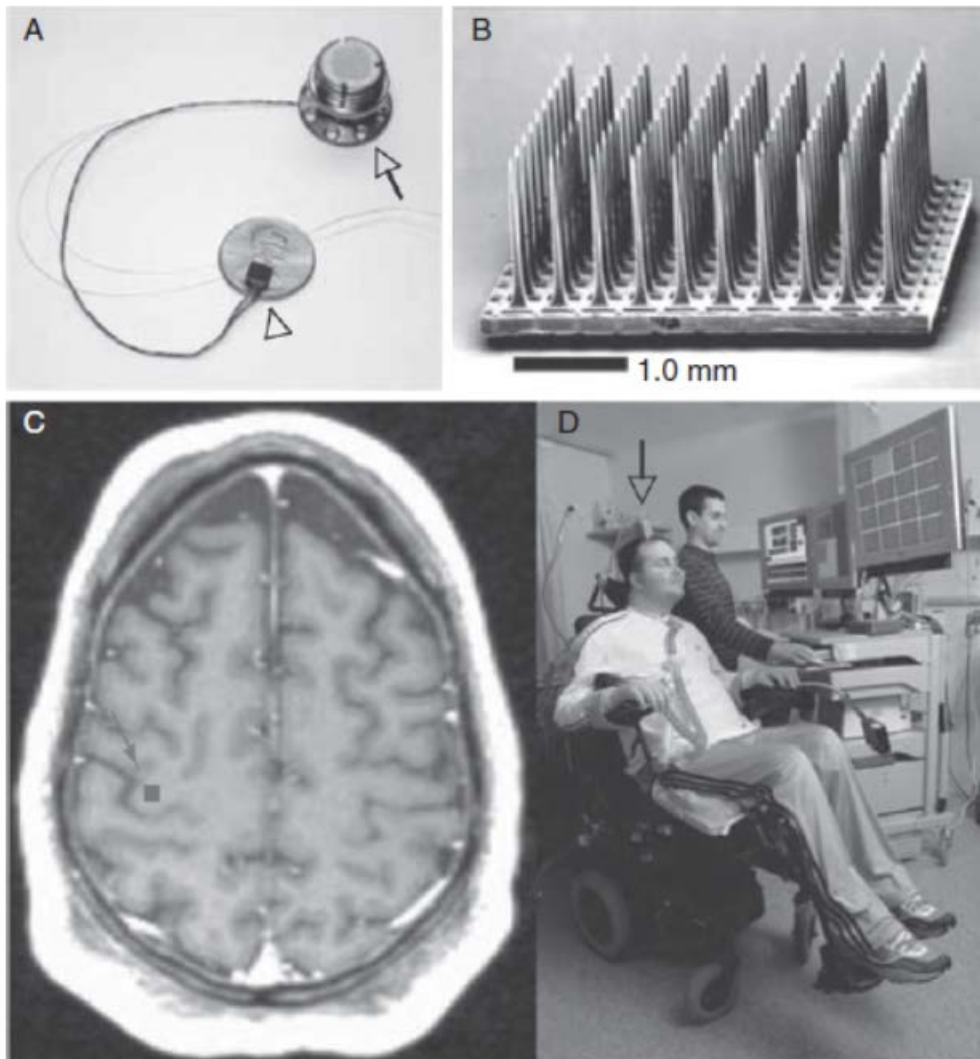
Brain control trial



B



Cognitive BCI for a reaching task.



Invasive BCI in a human.

Minghui Shi from Xiamen University

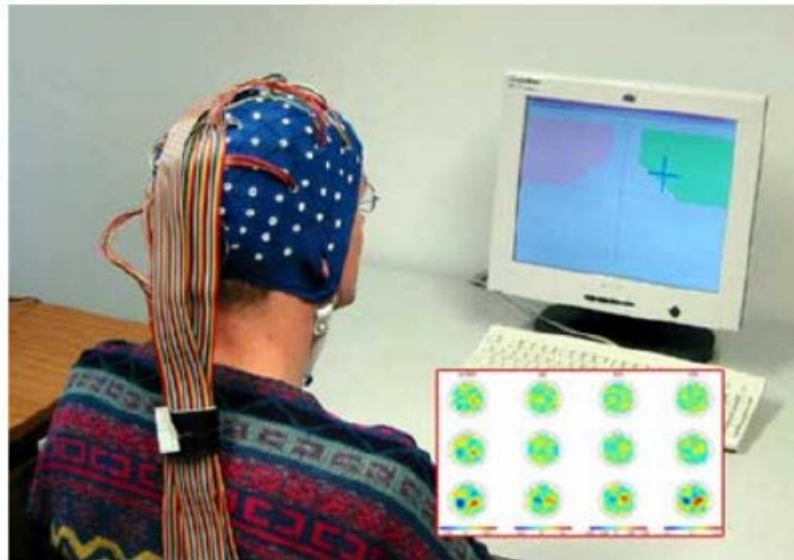
Figures of EEG-based BCIs



Berlin BCI:用于字符输入和真实游戏的BCI系统



Graz BCI:用于残疾人康复和虚拟现实环境的BCI系统



Minghui Shi from Xiamen University



清华大学BCI研究组一用于智能机器狗和电话拨号的BCI系统



华南理工大学BCI研究组一基于BCI的二维光标控制系统
和脑控轮椅

Textbook & References

- **Textbook:**

- Rao, R.P., ***Brain-computer interfacing: an introduction***. 2013: Cambridge University Press. ISBN 978-0-521-76941-9

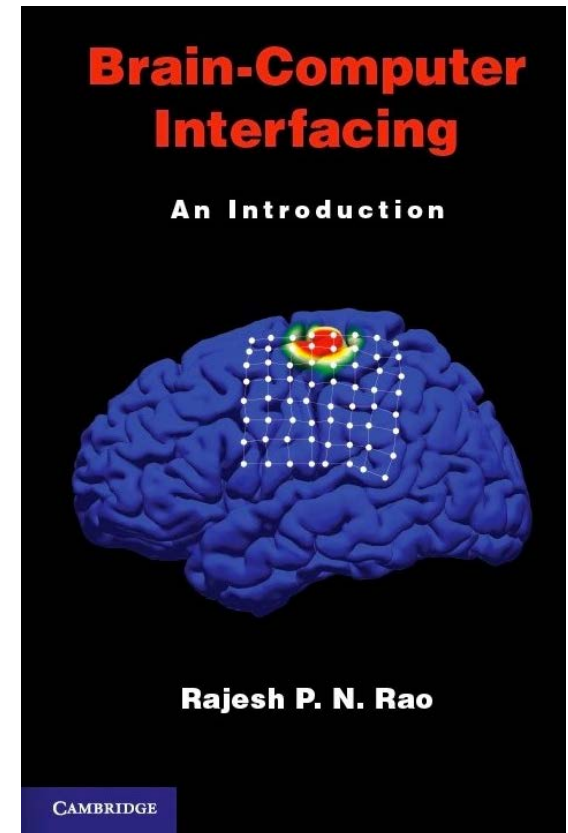
-

- **Reference:**

- M. Clerc, L. Bougrain, and F. Lotte, ***Brain Computer Interfaces: Fundamentals and Methods***: ISTE Ltd and John Wiley & Sons, Inc, 2016.
- Hassanien, A.E. and A.T. Azar, ***Brain-Computer Interfaces: Current Trends and Applications***. Vol. 74. 2014: Springer.
- Wolpaw, J. and E.W. Wolpaw, ***Brain-computer interfaces: principles and practice***. 2012: Oxford University Press.

Textbook

- **Key features** include:
 - ▣ Essential **background**.
 - ▣ Detailed description of the major **types** of BCIs.
 - ▣ In-depth discussion of BCI **applications** and BCI **ethics**.
 - ▣ <http://bci.cs.washington.edu>



Reference Book



脑机接口导论

作 者： [印度]拉杰什 P.N.拉奥 (Rajesh P. N. Rao) 著

译 者： 张莉 陈民铀 译

丛 书 名： 计算机科学丛书

出版日期： 2016-07-11

I S B N： 978-7-111-53995-7

定 价： 89.00元

Content & Topics

- This course focuses on the topic of **brain-computer interface (BCI)**, an emerging research on human-machine interaction in recent years.
- **The basic content include:**
 - basic neuroscience *background*
 - recording and analyzing **methods** for brain signals
 - fundamental *principle* of BCI
 - various BCI *applications*
 - *ethical* and *social* issues ensued from BCI

Content & Topics

- Specifically, the following **six topics** are emphasized in this course:
 1. What is BCI?
 2. What aspects of neuroscience research have made BCI possible?
 3. What are the main techniques of signal processing and machine learning used in BCI?
 4. What is the current state-of-the-art in BCIs?
 5. What limitations still need to be overcome to make BCIs more commonplace and useful for day-to-day use?
 6. What are the ethical, moral, and societal implications of BCIs?

Content & Topics

- **1. Introduction**
- **Part I Background**
- **Part II Putting It All Together**
- **Part III Major Types of BCIs**
- **Part IV Applications and Ethics**

Content & Topics

- **1. Introduction**
- **Part I Background**
 - ▣ **2. Basic Neuroscience**
 - ▣ **3. Recording and Stimulating the Brain**
 - ▣ **4. Signal Processing**
 - ▣ **5. Machine Learning**
- **Part II Putting It All Together**
- **Part III Major Types of BCIs**
- **Part IV Applications and Ethics**

Content & Topics

- **1. Introduction**
- **Part I Background**
- **Part II Putting It All Together**
 - ▣ **6. Building a BCI**
- **Part III Major Types of BCIs**
- **Part IV Applications and Ethics**

Content & Topics

- **1. Introduction**
- **Part I Background**
- **Part II Putting It All Together**
- **Part III Major Types of BCIs**
 - ▣ **7. Invasive BCIs**
 - ▣ **8. Semi-Invasive BCIs**
 - ▣ **9. Noninvasive BCIs**
 - ▣ **10. BCIs that Stimulate**
 - ▣ **11. Bidirectional and Recurrent BCIs**
- **Part IV Applications and Ethics**

Content & Topics

- **1. Introduction**
- **Part I Background**
- **Part II Putting It All Together**
- **Part III Major Types of BCIs**
- **Part IV Applications and Ethics**
 - ▣ **12. Applications of BCIs**
 - ▣ **13. Ethics of Brain-Computer Interface**

Basic Requirement

- ❑ Not for exam, but for knowledge.
- ❑ Do not miss any class.
- ❑ Positively participate in activities in each class.
- ❑ Search, read and discuss related references.

Group Task

- Each group has 2-4 students.
- Each group should make a presentation related to one of topics of BCI.
- Each group submit a comprehensive report.

Score Assessment



- General Performance (20%)
- Oral Presentation (40%)
- Written Report (40%)

本讲内容

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- **Content & Topics**
- **Basic Requirement**
- **Group Task**
- **Score Assessment**

End



- Thank you for your attention.
- Any questions?